

 AI ENGINEERING STACK

Building a Production RAG Pipeline with LangGraph

From raw PDFs to grounded answers: step by step,
with code.



PDF



Chunks



Vectors



Retrieve



Answer



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What is RAG, and why does it **matter**?

RAG (Retrieval-Augmented Generation) is a technique that grounds an LLM's answers in real, external data instead of relying only on what the model memorized during training.

✓ WHY IT'S IMPORTANT

- ✓ LLMs hallucinate when they don't know something: RAG gives them a source of truth to lean on
- ✓ No need to fine-tune a model every time your data changes
- ✓ You can cite sources, which builds trust
- ✓ Keeps answers current with your latest documents

📁 COMMON USE CASES:

📄 Internal company docs

🎧 Support product manuals

⚖️ Legal & policy Q&A

</> Code doc assistants



Why LangGraph instead of plain LangChain?

LangGraph is a library (built by the LangChain team) for building LLM applications as **stateful graphs** instead of linear chains.

LANGCHAIN (CHAINS)

- ✗ Runs mostly linearly: good for simple pipelines
- ✗ Awkward once you need loops, retries, or branching logic
- ✗ Harder to debug flow hidden inside chain composition

LANGGRAPH (GRAPHS)

- ✓ Models `node` and `edge` with app as shared state
- ✓ Native support for conditional branching & retries
- ✓ Built-in persistence & human-in-the-loop support
- ✓ Explicit graph makes debugging & visualization easy

WHAT WE NEED BEFORE BUILDING:

Python 3.10+

langgraph & langchain

sentence-transformers

faiss-cpu

pypdf

Open-source LLM



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What we're building today

An end-to-end RAG pipeline, orchestrated as a LangGraph graph:



1. PDF Loading



2. Chunking



3. Embeddings



4. FAISS Store



5. Retrieval



6. Reranking



7. Prompting



8. Generation



9. Evaluation

Dependencies to install:

```
pip install langgraph langchain langchain-community \
sentence-transformers faiss-cpu pypdf \
transformers accelerate ragas
```

✓ **pypdf:** PDF extraction

✓ **sentence-transformers:** embed + rerank

✓ **transformers:** open LLM

✓ **langchain:** text chunking

✓ **faiss-cpu:** vector search

✓ **ragas:** RAG evaluation



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Extracting text from PDFs

Reads raw PDF files and pulls out clean text so it can be processed downstream. This is the entry point of the pipeline: garbage in, garbage out, so clean extraction matters.

```
from langchain_community.document_loaders import PyPDFLoader

def load_pdf(file_path: str):
    loader = PyPDFLoader(file_path)
    documents = loader.load() # returns 1 Document per page
    return documents

docs = load_pdf("company_handbook.pdf")
print(f"Loaded {len(docs)} pages")
```

i KEY TAKEAWAYS

- **Page-by-page wrapping:** PyPDFLoader reads the PDF page-by-page and wraps each page in a Document object with `page_content` and `metadata`.
- **Why metadata matters:** Keeping metadata here matters: later, when the LLM cites an answer, you want to know which page it came from.
- **Scanned PDFs need OCR:** For scanned/image PDFs you'd add OCR (e.g., `unstructured` or `pytesseract`): plain text extraction won't work there.



Splitting documents into chunks

Breaks large documents into smaller, overlapping pieces so each chunk is small enough to embed meaningfully and fit into the LLM's context window later.

```
from langchain.text_splitter import RecursiveCharacterTextSplitter

def chunk_documents(documents, chunk_size=500, chunk_overlap=50):
    splitter = RecursiveCharacterTextSplitter(
        chunk_size=chunk_size, chunk_overlap=chunk_overlap,
        separators=["\n\n", "\n", ". ", " ", ""]
    )
    return splitter.split_documents(documents)

chunks = chunk_documents(docs)
print(f"Created {len(chunks)} chunks")
```

HOW IT WORKS

- > Semanti chunk_s focus:** keeps each piece small enough to be semantically focused (too big = diluted embeddings, too small = lost context).
- > Preserving boundaries:** chunk_overl prevents important info from being cut off exactly at a chunk boundary.
- > Natural boundaries first:** RecursiveCharacter tries to split on paragraphs and sentences first before falling back to words.



Turning text into vectors

Converts each text chunk into a numerical vector that captures its semantic meaning, so we can later compare "how similar" a question is to a chunk mathematically.

```
from sentence_transformers import SentenceTransformer

embedding_model = SentenceTransformer("BAAI/bge-small-en-v1.5")

def embed_chunks(chunks):
    texts = [chunk.page_content for chunk in chunks]
    return embedding_model.encode(
        texts, normalize_embeddings=True, show_progress_bar=True
    )

vectors = embed_chunks(chunks)
```

WHY THESE SETTINGS?

- > Open-source efficiency:** BAAI/bge-small-en-v1.5 is a strong, free, open-source embedding model: small enough to run on CPU, good enough for most tasks.
- > Reliable cosine search:** `normalize_embeddings=True` scales vectors to unit length, making cosine similarity search reliable.
- > Semantic vector space:** Each chunk becomes a fixed-length vector (384 dimensions): similar meanings end up close in vector space.



Storing vectors for fast search

Stores all chunk embeddings in FAISS, a library built for extremely fast similarity search over millions of vectors: this becomes our searchable knowledge base.

```
import faiss
import numpy as np

def build_faiss_index(vectors):
    dimension = vectors.shape[1]
    index = faiss.IndexFlatIP(dimension) # inner product = cosine
    index.add(np.array(vectors).astype("float32"))
    return index

faiss_index = build_faiss_index(vectors)
faiss.write_index(faiss_index, "knowledge_base.index")
```

FAISS ADVANTAGES

- > Inner product math:**

IndexF

does an exact inner-product search: since our embeddings are normalized, this equals cosine similarity.
- > Zero API cost:**

FAISS runs entirely locally (no API calls, no cost) and is fast enough for most use cases up to millions of vectors.
- > Disk persistence**

We persist the index to disk so we don't have to re-embed everything every time we run the app.



Finding relevant chunks for a query

Takes the user's question, embeds it the same way, and searches FAISS for the top-k most similar chunks.

```
def retrieve(query: str, index, chunks, top_k=5):
    query_vector = embedding_model.encode(
        [query], normalize_embeddings=True
    ).astype("float32")

    scores, indices = index.search(query_vector, top_k)
    return [chunks[i] for i in indices[0]]

results = retrieve("What is the leave policy?", faiss_index, chunks)
```

i RETRIEVAL RULES

- **Same model requirement:** The query must be embedded with the *same* model used for the chunks: otherwise the vector spaces won't line up.
- **Tuning k :** top_k controls how many candidate chunks we pull: larger k gives the reranker more to work with but costs more compute downstream.
- **Coarse filter:** This is "coarse" retrieval: fast but not perfectly precise, which is exactly why we rerank next.



Reranking for precision

Re-scores the retrieved chunks using a more accurate (but slower) cross-encoder model, so the most relevant chunks end up at the top before we hand them to the LLM.

```
from sentence_transformers import CrossEncoder

reranker = CrossEncoder("cross-encoder/ms-marco-MiniLM-L-6-v2")

def rerank(query, retrieved_chunks, top_n=3):
    pairs = [(query, c.page_content) for c in retrieved_chunks]
    scores = reranker.predict(pairs)
    ranked = sorted(
        zip(scores, retrieved_chunks), key=lambda x: x[0], reverse=True
    )
    return [chunk for _, chunk in ranked[:top_n]]

top_chunks = rerank("What is the leave policy?", results)
```

WHY RERANK?

- > Cross-encoder precision:** Vector search is fast but approximate; a cross-encoder looks at query and chunk *together* for higher relevance accuracy.
- > Efficiency balance:** We only rerank the small top-k set from retrieval (not the whole database): this keeps it fast while boosting precision.
- > Reducing noise:** `top_n=` trims down to just the best chunks, reducing noise in the final prompt.



Building the prompt

Combines the retrieved context and the user's question into a structured prompt that instructs the LLM to answer only from the provided context.

```
def build_prompt(query, top_chunks):
    context = "\n\n".join([c.page_content for c in top_chunks])
    return f"""Answer using ONLY the context below.
If not in context, say "I don't have enough information."

Context:
{context}

Question: {query}
Answer: ""

final_prompt = build_prompt("What is the leave policy?", top_chunks)
```

PROMPT ENGINEERING TIPS

- **Reducing hallucinations:** Explicitly telling the model to stick to the context is a simple but effective way to reduce hallucination.
- **Safe fallback:** The fallback instruction ("say I don't have enough information") prevents the model from making things up when retrieval comes up empty.
- **Wording consistency:** Keep this prompt clean and consistent: small wording changes can noticeably affect answer quality.



Generating the answer

Sends the constructed prompt to an open-source LLM to generate the final, grounded answer.

```
from transformers import pipeline

generator = pipeline(
    "text-generation",
    model="mistralai/Mistral-7B-Instruct-v0.2",
    device_map="auto"
)

def generate_answer(prompt, max_new_tokens=300):
    out = generator(prompt, max_new_tokens=max_new_tokens, do_sample=False)
    return out[0]["generated_text"][len(prompt):].strip()

print(generate_answer(final_prompt))
```

GENERATION SETTINGS

- > Strong instruction model:** Mistral-7B-Instruct is a solid open-source choice; runs well on GPU with strong instruction following.
- > Deterministic output:** `do_sample=False` makes output deterministic: ideal for factual RAG answers without creative variance.
- > Production scaling:** In production, swap for a hosted inference endpoint (HF Endpoints, vLLM, Ollama) for better latency.



Putting it all together as a LangGraph

Wraps every step into a LangGraph graph: functions become nodes, and the graph controls the flow (supporting retries & conditional branches).

```
from langgraph.graph import StateGraph, END
from typing import TypedDict, List

class RAGState(TypedDict):
    query: str; retrieved: List; reranked: List; prompt: str; answer: str

def retrieve_node(s): s["retrieved"] = retrieve(s["query"], faiss_index, chunks); return s
def rerank_node(s): s["reranked"] = rerank(s["query"], s["retrieved"]); return s
def prompt_node(s): s["prompt"] = build_prompt(s["query"], s["reranked"]); return s
def generate_node(s): s["answer"] = generate_answer(s["prompt"]); return s

graph = StateGraph(RAGState)
graph.add_node("retrieve", retrieve_node); graph.add_node("rerank", rerank_node)
graph.add_node("prompt", prompt_node); graph.add_node("generate", generate_node)
graph.set_entry_point("retrieve"); graph.add_edge("retrieve", "rerank")
graph.add_edge("rerank", "prompt"); graph.add_edge("prompt", "generate")
graph.add_edge("generate", END)

rag_app = graph.compile()
print(rag_app.invoke({"query": "What is the leave policy?"})["answer"])
```

- **Shared State:** RAGState is passed between nodes: every node reads and writes to it.
- **Modular Nodes:** Each step is an explicit node: easy to test or reorder independently.
- **Graph Power:** Easily add conditional edges (e.g., loop back if rerank score is low).



Evaluating the RAG pipeline

Measures whether the pipeline is actually retrieving relevant context and generating faithful, accurate answers using automated RAG evaluation metrics.

```
from ragas import evaluate
from ragas.metrics import faithfulness, answer_relevancy, context_precision
from datasets import Dataset

eval_data = Dataset.from_dict({
    "question": ["What is the leave policy?"],
    "answer": [result["answer"]],
    "contexts": [[c.page_content for c in result["reranked"]]],
    "ground_truth": ["Employees get 18 paid leave days per year."]
})

scores = evaluate(eval_data, metrics=[faithfulness, answer_relevancy,
context_precision])
print(scores)
```

i RAGAS METRICS

- **faithfulness:** Checks if answer is supported by context (catches hallucination).
- **answer_relevancy:** Checks if answer directly addresses the question asked.
- **context_precision** Checks if retrieved chunks were actually useful and relevant.
- **Data-driven tuning:** Run regularly after changes to measure real pipeline improvements.



🚩 CONCLUSION

That's a full RAG pipeline, end to end

All orchestrated as a clean, stateful LangGraph graph. Here is your complete architecture recap:

 1. PDF Loading →  2. Chunking →  3. Embeddings

 4. FAISS Store →  5. Retrieval →  6. Reranking

 7. Prompting →  8. Generation →  9. Evaluation



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